THE LANCET

Supplementary appendix

This appendix formed part of the original submission and has been peer reviewed. We post it as supplied by the authors.

Supplement to: Rugo HS, Bardia A, Marmé F, et al. Overall survival with sacituzumab govitecan in hormone receptor-positive and human epidermal growth factor receptor 2-negative metastatic breast cancer (TROPiCS-02): a randomised, open-label, multicentre, phase 3 trial. *Lancet* 2023; published online Aug 23. https://doi.org/10.1016/S0140-6736(23)01245-X.

Overall survival with sacituzumab govitecan in HR+/HER2- metastatic breast cancer from the randomised, phase 3 TROPiCS-02 trial

Rugo et al.

SUPPLEMENTARY APPENDIX

Table of Contents **Page Numbers**

List of TROPiCS-02 Study Principal Investigators **2**

Additional Methods **5**

Supplementary Figures **6**

Figure S1: Hierarchical testing procedure **6**

Figure S2: Kaplan-Meier Estimates of Time to Deterioration (TTD) of Global health status/QoL Scale **7**

Figure S3: Kaplan-Meier Estimates of Time to Deterioration (TTD) of Fatigue Scale **8**

Supplementary Tables **9**

Table S1: Representativeness of study participants **9**

Table S2. Patient demographics and baseline characteristics for patients by initial use of CDK4/6 in metastatic setting (1L vs ≥2L) **11**

Table S3: Patient demographics and baseline characteristics for patients with Trop-2 H-scores <100 and ≥100 **12**

Table S4. PFS and OS in patients by initial use of CDK4/6 in metastatic setting (1L vs ≥2L) **13**

Table S5. PFS and OS in patients with Trop-2 H-scores <100 and ≥100 **14**

Table S6: Summary of common treatment-related adverse events of any grade (≥10%) and by worst grade ≥3 (≥5%) in any treatment arm **15**

Table S7: Overall summary of treatment-emergent adverse events by UGT1A1 status **17**

Table S8: Treatment-emergent adverse events of special interest by UGT1A1 status **18**

References **19**

List of TROPiCS-02 Study Principal Investigators.

Study Investigator	Study Institution
Maysa Abu-Khalaf	Thomas Jefferson University
Foluso Ademuyiwa	Washington University School of Medicine
Clarence Adoo	HonorHealth Research Institute
Philippe Aftimos	Institut Jules Bordet
Samreen Ahmed	University Hospitals of Leicester NHS Trust
Cristina Alencar	Virginia Oncology Associates
Anne Armstrong	The Christie NHS Foundation Trust
Cristina Arqueros	Hospital de la Santa Creu i de Sant Pau
Jean-Pierre Ayoub	Centre Hospitalier De l'Universite De Montreal CHUM
Aditya Bardia*	Massachusetts General Hospital
J. Thaddeus Beck	Highlands Oncology Group
Giampaolo Bianchini	Ospedale San Raffaele
Sibel Blau	Northwest Medical Specialties, PLLC
Nathalie Bonnin	Hospices Civils de Lyon
Adam Brufsky	UPMC Hillman Cancer Center
Christina Brzezniak	Virginia Cancer Specialists, PC
Giuseppe Cairo	Ospedale Vito Fazzi di Lecce
Lourdes Calvo Martinez	Complejo Hospitalario Universitario A Coruña
Luigi Cavanna	Azienda Unità Sanitaria Locale di Piacenza-Ospedale Guglielmo da Saliceto
Marina Elena Cazzaniga	Azienda Ospedaliera San Gerardo di Monza
Eva Ciruelos	Hospital Universitario 12 de Octubre
Patrick Cobb	Saint Vincent Frontier Cancer Center
Javier Cortes Castan*	Hospital Ruber Internacional
Ricardo Costa	Moffitt Cancer Center
Florence Dalenc	Institut Claudius Regaud
Maaike de Boer	Maastricht University Medical Centre
Neelam Desai	Beth Israel Deaconess Medical Center
Jennifer Diamond	University of Colorado
Erion Dobi	Hôpital Jean Minjoz
Lara Durna	Kaiser Permanente Medical Center
Peter Fasching	Universitätsklinik Erlangen
Gianluigi Ferretti	Istituto Regina Elena
Nelly Firmin	Institut Regional du Cancer de Montpellier
Keerthi Gogineni	Emory University
Patricia Gomez Pardo	Hospital Universitari Vall D'Hebron
Julien Grenier	Institut Sainte Catherine
Erika Hamilton	Tennessee Oncology
Susanna Hegewisch-Becker	Onkologische Schwerpunktpraxis Eppendorf

Study Investigator	Study Institution
Stephanie Henry	CHU UCL NAMUR - Sainte Elisabeth
Dawn Hershman	Columbia University Medical Center
Claudine Isaacs	Georgetown University
Jean-Philippe Jacquin	Institut de Cancerologie Lucien Neuwirth
Komal Jhaveri	Memorial Sloan-Kettering Cancer Center
Virginia Kaklamani	University of Texas Health Science Center at San Antonio
Hans-Christian Kolberg	Marienhospital Bottrop
Amy Krie	Virginia Piper Cancer Center (Alliant Health)
Christian Kurbacher	Gynakologisches Zentrum Bonn
Charles Kurkul	Texas Oncology PA
Sylvain Ladoire	Centre Georges François Leclerc
Sabine Linn	Antoni van Leeuwenhoekziekenhuis
Delphine Loirat	Institut Curie
Rafael Lopez	Hospital Clínico Universitario de Santiago de Compostela - CHUS
Kristina Lübbe	DIAKOVERE Krankenhaus gGmbH Henriettenstift - Standort Kirchrode
Hans-Joachim Luck	Gynakologisch-Onkologische Praxis Hannover
Elisabeth Luporsi	Hôpital de Mercy
Ling Ma	Rocky Mountain Cancer Center
Rolf Mahlberg	Klinikum Mutterhaus der Borromaerinnen
Frederik Marme*	Universitatsmedizin Mannheim
Miguel Martin Jimenez	Hospital General Universitario Gregorio Marañón
Eduardo Martinez	Hospital Provincial de Castellón
Ingrid Mayer	Vanderbilt University
Kelly McCann	University of California Los Angeles (UCLA)
Serafín Morales	Hospital Universitari Arnau De Vilanova (Huav)
Sarah Mougalian	Yale University
Rita Nanda	University of Chicago
Polly Niravath	Houston Methodist Hospital
Yelena Novik	New York University School of Medicine
Shannon O'Connor	Maryland Oncology Hematology
Alicia Okines	The Royal Marsden NHS Foundation Trust
Coral Omene	Rutgers Cancer Institute of New Jersey
Rianne Oosterkamp	Medisch Centrum Haaglanden Antoniushove
Joyce O'Shaughnessy	Texas Oncology
Steven Papish	Summit Medical Group
Ritesh Parajuli	University of California Irvine (UCIMC)
Felicity Paterson	Royal Surrey County Hospital NHS Foundation Trust
Michel Pavic	Centre Hospitalier De l'Universite De Sherbrooke Hôpital Fleurimont
Rebecca Pedersini	Azienda Ospedaliera Spedali Civili di Brescia
Alejandra Perez	Sylvester Comprehensive Cancer Center

Study Investigator	Study Institution
Timothy Pluard	St Luke's Cancer Institute
David Potter	Masonic Cancer Center
Paola Pozzi	Ospedale di Desio
Kevin Punie	Universitaire Ziekenhuis Leuven
Fabricio Racca	Hospital Quironsalud Barcelona
Daniel Rayson	Queen Elizabeth II Health Sciences Center
Mattea Reinisch	Kliniken Essen-Mitte
Paul Richards	Blue Ridge Cancer Care
Elizabeth Riley	James Graham Brown Cancer Center
David Riseberg	Mercy Medical Center
Regan Rostorfer	University of Florida
Hope Rugo*	University of California San Francisco
Manuel Ruiz Borrego	Hospital Universitario Virgen del Rocio
Sagar Sardesai	Stefanie Spielman Comprehensive Breast Center
Peter Schmid*	Barts Health NHS Trust
Andreas Schneeweiss	Nationales Centrum für Tumorerkrankungen Heidelberg
Holger Schultz	HELIOS Klinikum Berlin-Buch
Lee Schwartzberg	West Clinic
Lasika Seneviratne	Los Angeles Hematology Oncology Medical Group
Priyanka Sharma	University of Kansas
Sasha Strain	Texas Oncology PA
Rachel Swart	Arizona Oncology Associates
Karen Tedesco	New York Oncology Hematology - Amsterdam Cancer Center
Hans Tesch	Centrum für Hämatologie und Onkologie Bethanien
Amy Tiersten	Mount Sinai
Sara Tolaney*	Dana-Farber Cancer Institute
Olivier Trédan	Centre Leon Berard
Nisha Unni	University of Texas Southwestern Medical Center
Thierry Velu	Chirec Cancer Institute
Hermann Voss	Stadtisches Klinikum Dessau
Grace Wang	Baptist Health - Miami Cancer Institute
Rudolf Weide	Praxisklinik für Hämatologie und Onkologie Koblenz
Duncan Wheatley	Royal Cornwall Hospital Trust
Kay Yeung	University of California San Diego
Amelia Zelnak	Northside Hospital Atlanta

***Denotes members of the TROPiCS-02 scientific steering committee.**

ADDITIONAL METHODS

Alpha spending strategy for OS

The Lan DeMets alpha spending function that approximates a Pocock (LD-Pocock) approach was used to account for multiplicity introduced by including OS interim analyses for superiority. Although the overall power from using LD-Pocock is relatively lower than that from using a Lan DeMets alpha spending function that approximates an O'Brien-Fleming approach (LD-OBF), LD-Pocock is the more favorable approach over LD-OBF, since LD-Pocock increases the probability of success at an earlier time point, which was able to provide as much as 70% power at the first interim analysis of OS. Given the compelling results from ASCENT for mTNBC,[1] LD-Pocock was chosen due to the expectation of a significant result at an earlier stage of the study.

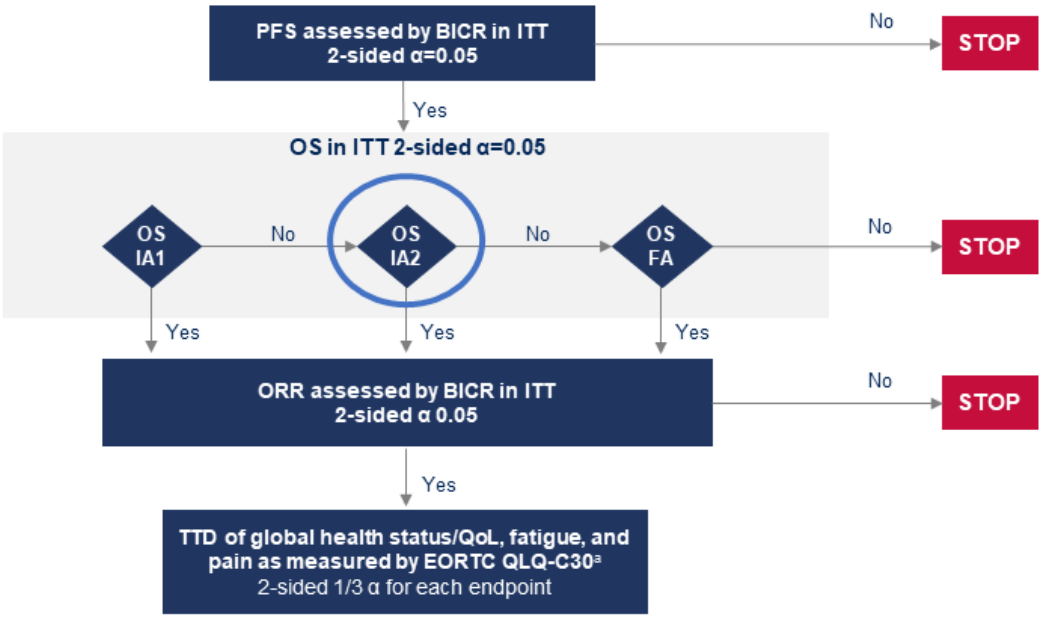
Major amendments

In the initial trial design for TROPiCS-02, ORR and PFS by local investigator assessment were dual primary endpoints. A protocol amendment was made to move ORR and PFS by local investigator assessment to secondary endpoints, and to add PFS by blinded independent central review as the only primary endpoint (prior to unblinding and without any access to, or knowledge of, efficacy data).

In protocol amendment 7, a descriptive non-comparative analysis of OS was planned at the time of the final primary analysis of PFS, when an administrative alpha of 0.00001 was planned to be spent. However, at the time of the final primary analysis of PFS, the targeted number of events required for first interim OS analysis had been reached (272 needed, 293 actual events). Therefore, the final analysis of PFS and the first interim superiority analysis of OS were conducted together, and the descriptive, non-comparative OS analysis was no longer applicable.

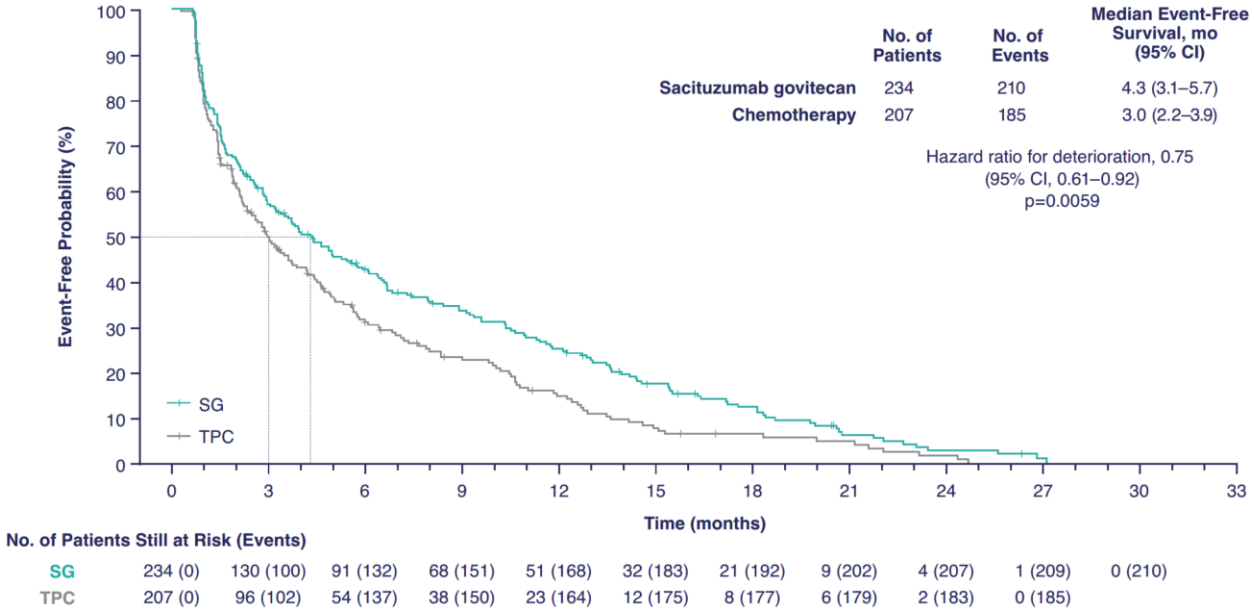
SUPPLEMENTARY FIGURES

Figure S1: Hierarchical testing procedure



[a]The 3 QoL endpoints will be tested using graphical approach of Maurer and Bretz to control multiplicity. BICR, blinded independent central review; EORTC QLQ-C30, European Organization for Research and Treatment of Cancer Quality of Life Questionnaire; FA, final analysis; IA, interim analysis; ITT, intent to treat; ORR, objective response rate; OS, overall survival; PFS, progression-free survival, QoL, quality of life; TTD, time to deterioration.

Figure S2: Kaplan-Meier Estimates of Time to Deterioration of Global Health Status/QoL Scale[a,b]

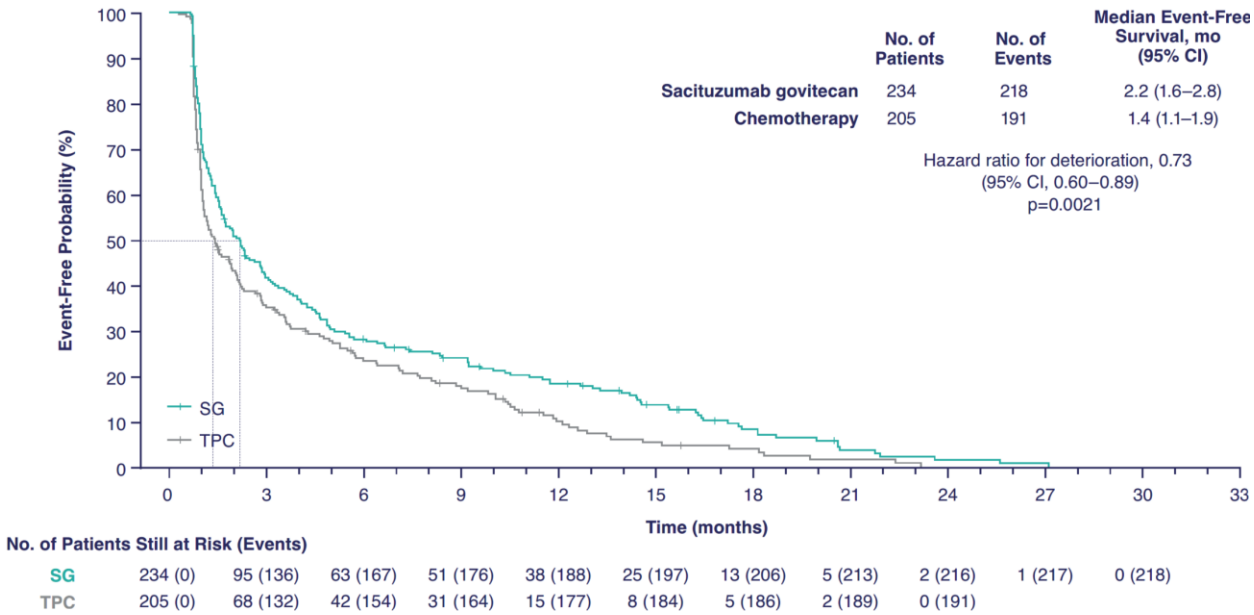


[a] Assessed in the EORTC QLQ-C30–evaluable population, defined as all patients in the intent-to-treat population with an evaluable EORTC QLQ-C30 assessment at baseline and at least 1 evaluable post-baseline assessment.

[b] Time to deterioration is defined as the time between randomisation and the time a patient experienced a ≥10-point worsening from baseline in a given domain.

EORTC QLQ-C30, European Organization for Research and Treatment of Cancer Quality of Life Questionnaire; SG, sacituzumab govitecan; TPC, treatment of physician's choice; QoL, quality of life.

Figure S3: Kaplan-Meier Estimates of Time to Deterioration of Fatigue Scale[a,b]



[a] Assessed in the EORTC QLQ-C30–evaluable population, defined as all patients in the intent-to-treat population with an evaluable EORTC QLQ-C30 assessment at baseline and at least 1 evaluable postbaseline assessment.
[b] Time to deterioration is defined as the time between randomisation and the time a patient experienced a ≥10-point worsening from baseline in a given domain.
EORTC QLQ-C30, European Organization for Research and Treatment of Cancer Quality of Life Questionnaire; SG, sacituzumab govitecan; TPC, treatment of physician's choice.

SUPPLEMENTARY TABLES

Table S1: Representativeness of study participants

Category	
Disease problem, or condition under investigation	HR-positive/HER2-negative advanced breast cancer
Special considerations related to	
Sex and gender	Breast cancer affects women more frequently than men, with 99% of cases occurring in women.[2]
Age	In the United States, the most common age group at diagnosis of HR-positive, HER2-negative breast cancers is 50 to 64 years. Approximately 60% of patients with HR-positive/HER2-negative breast cancers are <65 years at diagnosis.[3]
Race or ethnic group	Non-Hispanic White patients comprise 75% of HR-positive/HER2-negative breast cancer cases in the United States. Non-Hispanic Black, non-Hispanic Asian Pacific Islander, and Hispanic patients comprise 8%, 8%, and 9% of cases, respectively.[3]
Geography	Breast cancer incidence and mortality rates differ by country and geographic region. More developed countries/regions, including Australia and New Zealand, Europe, and North America, typically have the highest incidence rates of breast cancer.[4]
Other considerations	Most HR-positive/HER2-negative breast cancer cases are diagnosed in the localized (69%) and regional (26%) stages; only 4% of HR-positive/HER2-negative breast cancer cases are diagnosed in the distant (or metastatic) stage. As a result, exact population characteristics for advanced HR-positive/HER2-negative breast cancer are limited, but generally align with the characteristics of localized and regional stage cases.[5] Risk factors for HR-positive breast cancer include increased alcohol consumption, younger age at menarche, substantial weight gain in adult age, and use of postmenopausal hormone therapy.[6,7]
Overall representativeness of this trial	In this study, 229 (42%) patients were from North America and 314 (58%) were from Europe; 538 (99%) patients were female. In the total patient population, 362 (67%), 21 (4%), and 16 (3%) reported White, Black, and Asian race; collection of race and ethnicity data for 139 (26%) patients was not allowed by local regulators. Overall, the racial and ethnic

Category	
	characteristics in this study were representative of patients in the United States with HR-positive/HER2-negative breast cancer (all stages). It is unclear if Black and Asian women were underrepresented in this study due to the large number of patients who did not have race/ethnicity data collected. The median age of patients in this study was 56 years (IQR 49-65), and 403 (74%) patients were <65 years. This study population is slightly younger than the overall HR-positive/HER2-negative breast cancer population in the United States.

HER2, human epidermal growth factor receptor 2; HR, hormone receptor; IQR, interquartile range.

Table S2: Patient demographics and baseline characteristics for patients by initial use of CDK4/6 in metastatic setting (1L vs ≥2L)

	1L CDK4/6i (n=185)		**≥2L CDK4/6i (n=350)**	
	SG (n=84)	**TPC (n=101)**	**SG (n=183)**	**TPC (n=167)**
Median age, y (IQR)	55 (45-64)	54 (48-63)	59 (51-67)	56 (49-65)
Race, n (%)				
White	56 (67)	70 (69)	125 (68)	106 (64)
Non-White[a]	9 (11)	10 (10)	9 (5)	13 (8)
Not reported[b]	19 (23)	21 (21)	49 (27)	48 (29)
ECOG PS, n (%)				
0	37 (44)	47 (47)	76 (42)	78 (47)
1	47 (56)	54 (54)	107 (59)	89 (53)
Prior CDK4/6 inhibitor use, n (%)				
≤12 months	45 (54)	61 (60)	113 (62)	104 (62)
>12 months	36 (43)	39 (39)	68 (37)	62 (37)
Unknown	3 (4)	1 (1)	2 (1)	1 (1)
Prior endocrine therapy in metastatic setting ≥6 mo, n (%)	71 (85)	82 (81)	162 (89)	151 (90)
Visceral metastasis, n (%)				
Yes	83 (99)	97 (96)	171 (93)	158 (95)
No	1 (1)	4 (4)	12 (7)	9 (5)
Prior chemotherapy regimens for metastatic disease, n (%)				
2	47 (56)	50 50)	62 (34)	62 (37)
3-4	37 (44)	51 (51)	121 (66)	105 (63)

[a]Includes Black, Asian, American Indian or Alaska Native, Native Hawaiian or other Pacific Islander. [b]Not reported indicates local regulators did not allow collection of race or ethnicity information.
1L, first line; 2L, second line or greater; CDK4/6i, cyclin-dependent kinase 4/6 inhibitors; ECOG PS, Eastern Cooperative Oncology Group performance status, H-score, histochemical score; IQR, interquartile range; SG, sacituzumab govitecan; TPC, treatment of physician's choice.

Table S3: Patient demographics and baseline characteristics for patients with Trop-2 H-scores <100 and ≥100

	SG (n=238)		TPC (n=224)	
	H-Score <100[a] (n=96)	**H-Score ≥100 (n=142)**	**H-Score <100 (n=96)**	**H-Score ≥100 (n=128)**
Median age, y (IQR)	55 (48-65)	58 (50-64)	56 (48-65)	56 (48-65)
Race, n (%)				
White	65 (68)	97 (68)	64 (67)	87 (68)
Non-White[b]	8 (8)	7 (5)	8 (8)	8 (6)
Not reported[c]	23 (24)	38 (27)	24 (25)	33 (26)
ECOG PS, n (%)				
0	36 (38)	69 (47)	49 (51)	50 (39)
1	60 (63)	73 (51)	47 (49)	78 (61)
Prior CDK4/6 inhibitor use, n (%)				
≤12 months	53 (55)	91 (64)	57 (59)	86 (67)
>12 months	43 (45)	46 (32)	38 (40)	40 (31)
Unknown	0	5 (4)	1 (1)	2 (2)
Prior endocrine therapy in metastatic setting ≥6 mo, n (%)	82 (85)	125 (88)	81 (84)	114 (89)
Visceral metastasis, n (%)				
Yes	92 (96)	135 (95)	91 (95)	122 (95)
No	4 (4)	7 (5)	5 (5)	6 (5)
Prior chemotherapy regimens for metastatic disease, n (%)				
2	33 (34)	68 (48)	40 (42)	57 (45)
3-4	63 (66)	74 (52)	56 (58)	71 (55)

[a]Also includes a subset of patients with Trop ≤10 expression. [b]Includes Black, Asian, American Indian or Alaska Native, Native Hawaiian or other Pacific Islander. [c]Not reported indicates local regulators did not allow collection of race or ethnicity information.
CDK, cyclin-dependent kinase; ECOG PS, Eastern Cooperative Oncology Group performance status, H-score, histochemical score; IQR, interquartile range; SG, sacituzumab govitecan; TPC, treatment of physician's choice.

Table S4. PFS and OS per BICR in patients by initial use of CDK4/6 in metastatic setting (1L vs ≥2L)

	1L CDK4/6i (n=185)		**≥2L CDK4/6i (n=350)**	
	SG (n=84)	**TPC (n=101)**	**SG (n=183)**	**TPC (n=167)**
Median progression-free survival, months (95% CI)[a-c]	4.6 (3.1-6.4)	3.1 (1.7-4.4)	5.8 (4.4-7.3)	4.2 (3.2-5.4)
Hazard ratio (95% CI)[d]	0.64 (0.43-0.94)		0.66 (0.51-0.87)	
Overall survival events, no.	67	74	121	123
Median overall survival, months (95% CI)[e]	13.1 (10.5-15.4)	10.1 (8.6-13.5)	15.3 (13.3-18.1)	12.1 (10.5-13.5)
Hazard ratio (95% CI)[d]	0.92 (0.66-1.28)		0.75 (0.58-0.97)	

[a]PFS is defined as the number of months (30 days) from the date of randomisation to the date of the first radiological disease progression per RECIST 1.1 or death due to any cause, whichever comes first. [b]CI for median is computed using the Brookmeyer-Crowley method. [c]Median PFS and OS are from Kaplan-Meier estimates. [d]Hazard ratio is from unstratified Cox regression analysis. [e]OS is defined as the number of months (30 days) from date of randomisation to the date of death from any cause. Patients without documentation of death are censored on the date they were last known to be alive.
1L, first line; 2L, second line or greater; CDK4/6i, cyclin-dependent kinase 4/6 inhibitors; OS, overall survival; PFS, progression-free survival; RECIST, response evaluation criteria for solid tumors; SG, sacituzumab govitecan; TPC, treatment of physician's choice.

Table S5. PFS and OS in patients with Trop-2 H-scores <100 and ≥100

	H-Score <100[a]		**H-Score ≥100**	
	SG (n=96)	**TPC (n=96)**	**SG (n=142)**	**TPC (n=128)**
Median progression-free survival, months (95% CI)[b-d]	5.3 (4.1-6.0)	4.0 (2.8-5.6)	6.4 (4.0-8.3)	4.1 (2.1-4.5)
Hazard ratio (95% CI)[e]	0.77 (0.54-1.09)		0.60 (0.44-0.81)	
Overall survival events, no.	67	76	99	88
Median overall survival, months (95% CI)[f]	14.6 (12.7-18.1)	11.3 (10.0-13.3)	14.4 (12.7-16.4)	11.2 (9.9-12.9)
Hazard ratio (95% CI)[e]	0.75 (0.54-1.04)		0.83 (0.62-1.11)	

[a]Also includes a subset of patients with Trop ≤10 expression. [b]PFS is defined as the number of months (30 days) from the date of randomisation to the date of the first radiological disease progression per RECIST 1.1 or death due to any cause, whichever comes first. [c]CI for median is computed using the Brookmeyer-Crowley method. [d]Median PFS and OS are from Kaplan-Meier estimates. [e]Hazard ratio is from unstratified Cox regression analysis. [f]OS is defined as the number of months (30 days) from date of randomisation to the date of death from any cause. Patients without documentation of death are censored on the date they were last known to be alive.
H-score, histochemical score; OS, overall survival; PFS, progression-free survival; SG, sacituzumab govitecan; TPC, treatment of physician's choice; Trop-2, trophoblast cell-surface antigen 2.

Table S6: Summary of common treatment-related adverse events of any grade (≥10%) and by worst grade ≥3 (≥5%) in any treatment arm

		Sacituzumab govitecan (n=268)		**Chemotherapy (n=249)**	
Treatment-related adverse events, n (%)[a]		**All grade**	**Grade ≥3**	**All grade**	**Grade ≥3**
Hematologic	Neutropenia[b]	188 (70)	136 (51)	134 (54)	95 (38)
	Anaemia[c]	91 (34)	17 (6)	62 (25)	8 (3)
	Leukopenia[d]	37 (14)	23 (9)	23 (9)	13 (5)
	Lymphopenia[e]	31 (12)	10 (4)	25 (10)	8 (3)
	Thrombocytopenia[f]	17 (6)	1 (<1)	41 (17)	9 (4)
	Febrile neutropenia	14 (5)	14 (5)	11 (4)	11(4)
Gastrointestinal	Diarrhoea	152 (57)	25 (9)	42 (17)	3 (1)
	Nausea	148 (55)	3 (1)	77 (31)	7 (3)
	Vomiting	51 (19)	1 (<1)	30 (12)	4 (2)
	Constipation	50 (19)	0	36 (15)	0
	Abdominal pain	34 (13)	2 (1)	17 (7)	0
Investigations	AST increased	11 (4)	0	28 (11)	3 (1)
	ALT increased	12 (5)	0	24 (10)	6 (2)
Other	Alopecia	123 (46)	0	41 (17)	0
	Fatigue	101 (38)	15 (6)	73 (29)	7 (3)
	Asthenia	53 (20)	5 (2)	37 (15)	2 (1)
	Decreased appetite	42 (16)	1 (<1)	34 (14)	1 (<1)
	Neuropathy[g]	24 (9)	3 (1)	39 (16)	6 (2)

Assessed in the safety population.
[a]Patients may report more than one event per preferred term. Adverse events were coded using MedDRA v25.0, and adverse event severity was graded per NCI CTCAE v5.0.
[b]Combined preferred terms of 'neutropenia' and 'neutrophil count decreased.'
[c]Combined preferred terms of 'anaemia,' 'haemoglobin decreased,' and 'red blood cell count decreased.'
[d]Combined preferred terms of 'leukopenia' and 'white blood cell count decreased.'
[e]Combined preferred terms of 'lymphopenia' and 'lymphocyte count decreased.'

[f]Combined preferred terms of 'thrombocytopenia' and 'platelet count decreased.'
[g]Combined preferred terms of 'gait disturbance,' 'hypoesthesia,' 'muscular weakness,' 'neuropathy peripheral,' 'paraesthesia', and 'peripheral sensory neuropathy.'
ALT, alanine aminotransferase; AST, aspartate aminotransferase; MedDRA, Medical Dictionary for Regulatory Activities; NCI CTCAE, National Cancer Institute Common Terminology Criteria for Adverse Events.

Table S7: Overall summary of treatment-emergent adverse events by *UGT1A1* status

	Sacituzumab govitecan[a] (n = 268)		
Treatment-emergent adverse event, no. (%)[b]	***1/*1 Wild-type (n = 103)**	***1/*28 Heterozygous (n = 119)**	***28/*28 Homozygous (n = 25)**
Any grade	103 (100)	119 (100)	25 (100)
Grade ≥ 3	69 (67)	89 (75)	23 (92)
Leading to dose reduction, n (%)	26 (25)	49 (41)	10 (40)
Leading to treatment interruption, n (%)	70 (68)	76 (64)	19 (76)
Leading to treatment discontinuation, n (%)	5 (5)	7 (6)	3 (12)

[a]3 patients (1%) treated with SG had other genotypes, one each of *1/*36, *1/*37, and *28/*36.
[b]Treatment-emergent adverse events were defined as any adverse events that started on or after first dose date and up to 30 days after last dose date. Severity grades were defined using NCI CTCAE v5.0.
NCI CTCAE, National Cancer Institute Common Terminology Criteria for Adverse Events.

Table S8: Treatment-emergent adverse events of special interest by *UGT1A1* status

	Sacituzumab govitecan (n = 268)					
	***1/*1 Wild-type (n = 103)**		***1/*28 Heterozygous (n = 119)**		***28/*28 Homozygous (n = 25)**	
	Any grade	**Grade ≥ 3**	**Any grade**	**Grade ≥ 3**	**Any grade**	**Grade ≥ 3**
Treatment-emergent adverse events of special interest, no. (%)						
Neutropenia[a]	73 (71)	46 (45)	86 (72)	68 (57)	19 (76)	16 (64)
Diarrhea	60 (58)	6 (6)	77 (65)	15 (13)	17 (68)	6 (24)
Anemia[b]	34 (33)	6 (6)	43 (36)	10 (8)	12 (48)	2 (8)
Febrile neutropenia	6 (6)	6 (6)	8 (7)	8 (7)	1 (4)	1 (4)

[a]Combined preferred terms of 'neutropenia' and 'neutrophil count decreased.'
[b]Combined preferred terms of 'anemia,' 'hemoglobin decreased,' and 'red blood cell count decreased.'

REFERENCES

1. Bardia A, Hurvitz SA, Tolaney SM, et al. Sacituzumab govitecan in metastatic triple-negative breast cancer. N Engl J Med 2021;384(16):1529-1541.
2. World Health Organization Breast Cancer Fact Sheet. Available from: https://www.who.int/news-room/factsheets/detail/breast-cancer. Accessed October 5, 2022.
3. Howlader N, Altekruse SF, Li CI, et al. US incidence of breast cancer subtypes defined by joint hormone receptor and HER2 status. J Natl Cancer Inst 2014;106(5):dju055.
4. World Health Organization Globocan Breast Cancer Fact Sheet 2020. Available from: https://gco.iarc.fr/today/data/factsheets/cancers/20-Breast-fact-sheet.pdf. Accessed October 5, 2002.
5. Surveillance, Epidemiology, and End Results (SEER) Program SEER*Explorer Database: HR+/HER2– Breast Cancer Stage Distribution of SEER Incidence Cases, 2010-2019. Available from: www.seer.cancer.gov. Accessed October 5, 2022.
6. Ma H, Bernstein L, Ross RK, et al. Hormone-related risk factors for breast cancer in women under age 50 years by estrogen and progesterone receptor status: results from a case-control and a case-case comparison. Breast Cancer Res 2006;8(4):R39.
7. Rosenberg LU, Einarsdóttir K, Friman EI, et al. Risk factors for hormone receptor-defined breast cancer in postmenopausal women. Cancer Epidemiol Biomarkers Prev 2006;15(12):2482–88.